

SAFORA: A Community-Powered Safety Navigation & Emergency Alert Application

Synopsis

Submitted to

Dev Bhoomi Uttarakhand University

In partial fulfilment of the requirements for the degree of

Bachelor of Technology

in

Computer Science & Engineering

By

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DEV BHOOMI UTTARAKHAND UNIVERSITY

(Established by Govt. of Uttarakhand vide Dev Bhoomi Uttarakhand University Act No. 17 of 2021)

Batch (2026 - 2027)

CERTIFICATE

This is to certify that Ms. Arushi Saxena, Mr. Anurag Suyal, Mr. Aman Singh Kunwar, and Mr. Shubham Kumar are Bachelor of Technology students at Dev Bhoomi Uttarakhand University, Dehradun. They will work on the topic entitled "SAFORA: A Community-Powered Safety Navigation & Emergency Alert Application" under my supervision and guidance at the School of Engineering and Computing, Computer Science & Engineering, for the partial fulfilment of the requirements for the degree of Bachelor of Technology (B.Tech.) in Computer Science & Engineering.

Supervisor

Signature:

Name: Mr. Mukesh Rajput

Date:

DECLARATION BY RESEARCH SCHOLAR

We hereby declare that the synopsis of proposed project work entitled "SAFORA: A Community-Powered Safety Navigation & Emergency Alert Application" submitted in partial fulfilment of the requirements for the award of the degree of B.Tech. in Computer Science & Engineering and submitted to Dev Bhoomi Uttarakhand University, Dehradun, Uttarakhand, will be carried out by us under the expert guidance as well as direct supervision of Mr. Mukesh Rajput. We further declare that the work carried out by us shall be our original work and material obtained from other sources has been duly acknowledged in the synopsis.

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Title of the proposed Minor Project-I work:

"SAFORA: A Community-Powered Safety Navigation & Emergency Alert Application"

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1. Introduction

Personal safety while navigating cities, campuses, and neighbourhoods is a growing concern, particularly for women and students who frequently travel alone, at night, or through unfamiliar areas. While digital navigation has become ubiquitous, existing tools focus exclusively on speed and distance, leaving a significant gap in how people assess and manage safety-related risk during everyday travel. Safora is proposed as a mobile application that brings community-driven safety awareness and emergency communication into a single, integrated platform, allowing users to make safer travel decisions and stay connected to trusted contacts during a journey.

1.1 Problem Statement

Standard navigation applications such as Google Maps optimize purely for the fastest route, without accounting for safety factors such as lighting, crowd density, or past incident history. This leaves users, especially women and students walking alone at night or in unfamiliar areas, without any reliable way to judge whether a route is actually safe. There is currently no widely available, integrated mobile tool that combines community-verified safety information with real-time journey tracking and emergency alerting in a single application.

1.2 Objectives

1. To design and develop a mobile application that allows users to report unsafe locations and hazards.
2. To compute and visualize a community-generated safety score and heatmap for reported areas.
3. To implement a Safe Walk mode enabling users to share live location with trusted contacts during a journey, with automatic deviation and arrival-timeout alerts.
4. To implement a One-Tap SOS feature for immediate emergency alerts to trusted contacts.
5. To design an accurate geospatial data model using PostgreSQL with the PostGIS extension for real-world, distance-based queries.
6. To provide an administrative dashboard for moderating community reports and monitoring safety trends.

1.3 Scope of the Project

Application Area: Urban and campus personal-safety navigation and community hazard reporting.

Target Users: Students, women, and the general public who wish to make informed, safety-aware travel decisions and stay connected to trusted contacts while travelling.

Major Functions: Community hazard reporting, safety heatmap visualization, Safe Walk journey tracking, One-Tap SOS, trusted-contact management, and administrative moderation.

Boundaries: This Minor Project-I is scoped to a fully functional mobile application (Android and iOS) operating over standard internet connectivity. AI-based safe-route scoring, offline (Bluetooth/Wi-Fi Direct) emergency communication, and AI-based image detection for hazard classification are outside the boundary of this project and are identified as future scope.

Scalability: The data model and system architecture are designed so that advanced capabilities identified as future scope can be added in a subsequent development phase without redesigning the core system.

2. Literature Review / Existing System

Existing navigation applications such as Google Maps and Waze provide efficient routing based on distance and traffic conditions but do not incorporate any safety-related factors into their route recommendations, offering no way for a user to judge how safe a given path is.

SafetiPin is a crowd-sourced urban safety audit platform that allows users and volunteers to rate locations on parameters such as lighting and openness, producing safety scores for city areas. It is primarily an assessment and planning tool for city authorities and does not offer integrated real-time journey tracking or emergency alerting for individual users.

bSafe and similar personal-safety applications allow a user to trigger an alarm or begin live-streaming their location and surroundings to selected contacts in an emergency.

These applications focus on the emergency-response moment itself but do not provide any prior information about the relative safety of a chosen route.

Life360 and comparable family-locator applications allow continuous location sharing among a group of trusted contacts but do not incorporate any community-reported hazard data or safety scoring, and are oriented toward general location visibility rather than journey-specific safety monitoring.

3. Research / Development Gap

The reviewed systems each address a single aspect of the personal-safety problem in isolation: route calculation without safety context, safety assessment without real-time journey monitoring, or emergency alerting without any prior safety awareness. No widely available application integrates community-verified, geospatially accurate safety information with live journey tracking, automatic deviation handling, and one-tap emergency alerting within a single mobile platform. Safora is proposed to address this integration gap.

4. Proposed System

Safora is proposed as an integrated mobile application in which users report hazards and unsafe locations, the system aggregates these reports into a live, decay-weighted safety score visualized as a heatmap, and users can activate Safe Walk mode to share their journey with trusted contacts, with automatic warnings on route deviation and a One-Tap SOS available at all times. An administrative dashboard supports moderation of community reports to maintain data quality.

4.1 System Architecture / Flow Diagram

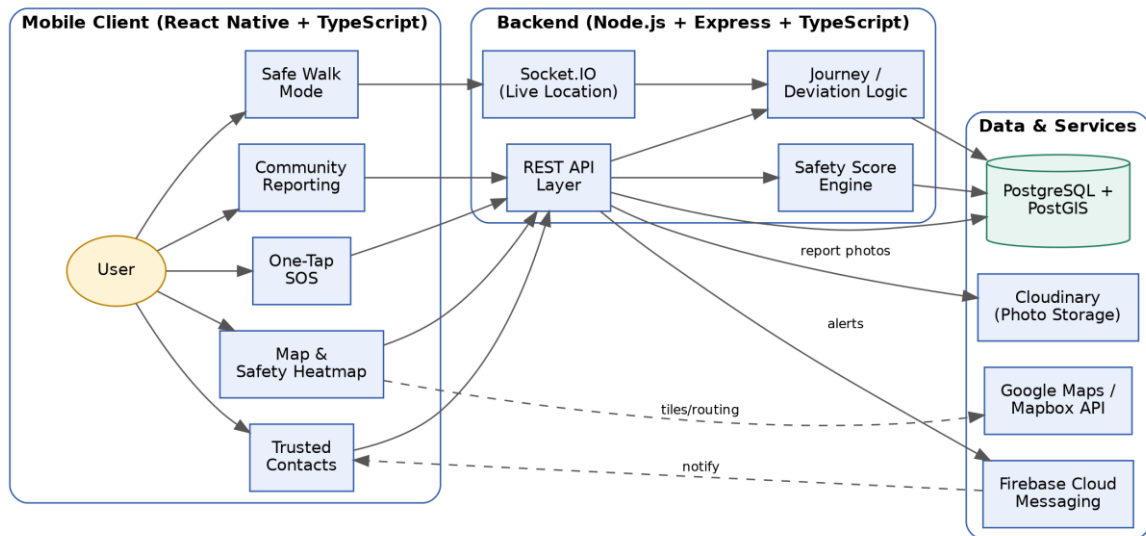


Figure 1: Proposed system architecture of Safora, showing the mobile client, backend API layer, and supporting data/services.

4.2 Modules of the Project

- Module 1: User Authentication & Profile Management
- Module 2: Community Hazard Reporting
- Module 3: Safety Score & Heatmap Engine
- Module 4: Safe Walk Mode & Trusted Contacts
- Module 5: One-Tap SOS Alert System
- Module 6: Administrative Dashboard & Moderation

4.3 Technologies and Tools

- Mobile Application: React Native with TypeScript
- Backend: Node.js with Express.js and TypeScript
- Database: PostgreSQL with the PostGIS extension for geospatial data
- ORM: TypeORM / Knex with native PostGIS spatial query support
- Authentication: JSON Web Tokens (JWT)
- Real-time Communication: Socket.IO
- Notifications: Firebase Cloud Messaging (FCM)

- File Storage: Cloudinary (report photo attachments)
- Maps: Google Maps SDK / Mapbox
- Background Location: react-native-background-geolocation
- Version Control: Git and GitHub

4.4 System Requirements

Sl. No	Requirement	Specification
1.	Development Machine (RAM)	8 GB or higher
2.	Storage	256 GB SSD or higher
3.	Test Devices	Android device and iOS device
4.	Operating System	Windows/macOS/Linux (development); Android 10+/iOS 14+ (target)
5.	Database	PostgreSQL 15.x with PostGIS 3.x
6.	Other Dependencies	Node.js 20.x, npm/pnpm, React Native CLI, Android Studio / Xcode

5. Expected Outcomes

The completed Minor Project-I is expected to deliver a fully functional mobile application in which community-reported hazards produce an accurate, continuously updated safety heatmap; Safe Walk mode reliably tracks a journey and issues timely deviation and arrival-timeout alerts without generating false alarms; One-Tap SOS delivers emergency alerts to trusted contacts promptly; and the administrative dashboard enables effective moderation of community-submitted reports.

5.1 Innovation / Novelty

The novelty of Safora lies in integrating three elements that existing applications address separately: a decay-weighted, geospatially clustered community safety score; a confirm-before-escalate deviation-handling flow for Safe Walk mode that reduces false alarms compared to naive deviation detection; and a PostGIS-backed geospatial data model designed from the outset to support the more advanced AI-driven and offline-communication capabilities identified as future scope.

6. Work Plan and Timeline

Activity	Month 1	Month 2	Month 3	Month 4	Status
Problem Identification	✓				Completed
Literature Review	✓				Completed
Requirement Analysis & System Design	✓				Planned
Development — Auth, Reporting, Safety Score		✓			Planned
Development — Safe Walk, SOS, Trusted Contacts		✓	✓		Planned
Admin Dashboard Development			✓		Planned
Testing & Evaluation			✓	✓	Planned
Documentation				✓	Planned
Final Presentation				✓	Planned

7. Conclusion

Safora addresses a genuine and persistent gap in personal-safety navigation by integrating community-verified hazard reporting, a live safety heatmap, Safe Walk journey tracking, and one-tap emergency alerting within a single mobile application. This Minor Project-I delivers a fully functional, real-world-usable system built on React Native, TypeScript, and PostgreSQL with PostGIS, providing a solid and extensible foundation for the application.

Future Scope: AI-based safe-route scoring using lighting and crowd-density data, offline Bluetooth/Wi-Fi Direct emergency communication for use during network outages, and AI-based image detection for automatic hazard classification are identified as valuable extensions beyond the scope of this Minor Project-I, and are proposed for a subsequent development phase.

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